SCHEME OF VALUATION- QP Code:25900199

MGU-UGP (HONOURS) EXAMINATION MARCH 2025

SECOND SEMESTER

Multi-Disciplinary Course (MDC) – MG2MDCSTA100-Time Series Methods and their Applications

Part A (Multiple Choice Questions)

(Answer **All** questions**.** Each question carries **2** marks**)**

1. (a) A model where components are multiplied
2. (b) To forecast future values based on past data
3. (b) The statistical properties remain constant over time
4. (c) To confirm that no additional patterns remain in the data
5. (a) A random walk depends only on the previous values, while an AR process depends on multiple past values
6. (a) Positive correlation
7. (a) An AR(1) process
8. (d) All of the above
9. (a) Autoregressive
10. (c) pacf()

Part B (Short Essay Type Questions)

(Answer any **three** questions. Each question carries **five** marks)

11. Explanation of trend---- 2.5 marks, Explanation of seasonality ---- 2.5 marks
12. Stationary time series ----2.5 marks, non-stationary time series ----2.5 marks
13. Month plot----2.5 marks, season plot---- 2.5 marks (explanation/use)
14. Correlation(explanation/formula of r)----2.5 marks, covariance(explanation/formula)----2.5 marks
15. Any two stationary processes----5 marks (2.5 marks each)